SL12-18 (12V 18Ah)

SL12-18 VRLA Deep Cycle battery provides superior high integrity and reliability. It is specially designed for frequent cycling charge and discharge. By using strong grids, thick plate and special active material are designed for repeated deep-discharge applications. It is designed for up to 6 years in standby service at 25 degree celcius or more than 300 cycles at 100% discharge in cycle service. Absorbent Glass Mat (AGM) technology type, valve regulated lead-acid (VRLA) battery no need to add distilled water or perform maintenance (Maintenance Free).



Battery case: Flame-retardant

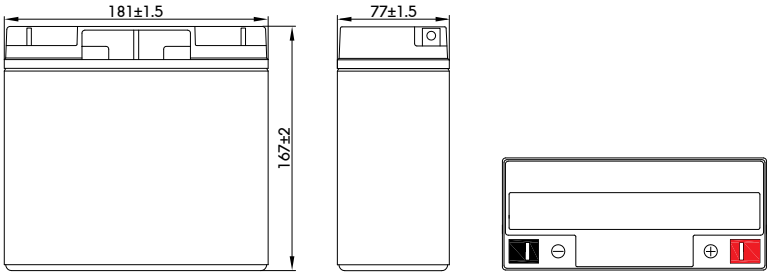
Specification

Cells Per Unit	6
Voltage Per Unit	12
Capacity	18Ah @20hr-rate to 1.75V per cell @25°C
Weight	Approx. 5.60 kg
Max. Discharge Current	180A (5sec)
Internal Resistance	≤11.0mΩ, fully charged @25°C
Operating Temperature Range	Discharge: -15°C~50°C Charge: -15°C~40°C Storage: -15°C~40°C
Nominal Operating Temperature Range	25°C±3°C
Float Charging Voltage	13.6 to 13.8VDC per unit @25°C
Recommended Max. Charging Current Limit	5.4A
Equalization and Cycle Service	14.5 to 14.9VDC per unit @25°C
Self Discharge	Battery can be stored for more than 6 months at 25°C, self-discharge rate less than 3% per month. Charging before use is recommended. For higher temperature, the time interval will be shorter.
Terminal	L Terminal to accept M5 nut & bolt (T3)
Container Material	Flame retardant ABS grade UL94 V-0

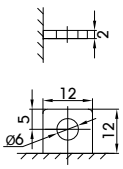
Dimension

Dimension: 181 (L) x 77 (W) x 167 (H) x 167 (TH)

Unit: mm



Terminal



Constant Current Discharge Characteristics : A (25°C)

F.V/Time	5min	15min	30min	1h	3h	5h	10h	20h
1.60V/Cell	67.5	37.9	21.76	12.03	5.18	3.48	1.951	1.046
1.67V/Cell	66.1	37.2	21.60	11.91	5.16	3.46	1.941	1.041
1.70V/Cell	64.2	36.1	21.33	11.74	5.12	3.42	1.933	1.035
1.75V/Cell	60.8	34.4	20.86	11.44	5.06	3.35	1.917	1.025
1.80V/Cell	54.5	31.7	19.99	10.87	4.89	3.23	1.859	1.002

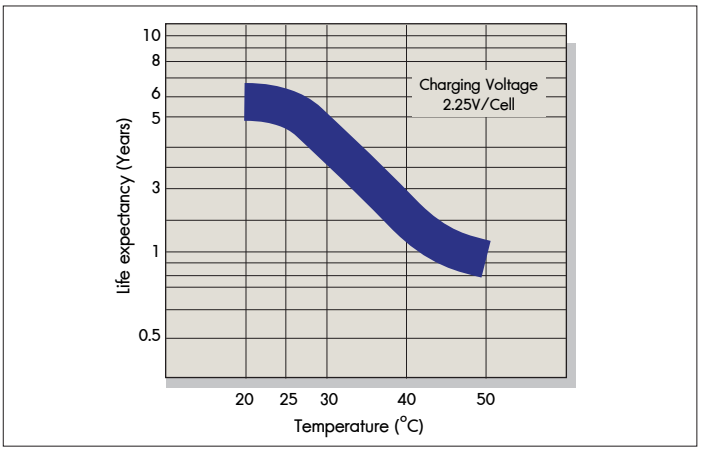
Constant Power Discharge Characteristics : W/Cell (25°C)

F.V/Time	5min	15min	30min	1h	3h	5h	10h	20h
1.60V/Cell	114.0	72.8	42.04	22.86	10.21	6.80	3.84	2.075
1.67V/Cell	109.0	69.0	41.78	22.66	10.17	6.77	3.82	2.070
1.70V/Cell	102.0	64.5	41.39	22.40	10.08	6.71	3.80	2.059
1.75V/Cell	92.1	59.0	40.58	22.00	9.96	6.60	3.76	2.035
1.80V/Cell	78.9	52.2	38.94	21.27	9.63	6.36	3.64	1.991

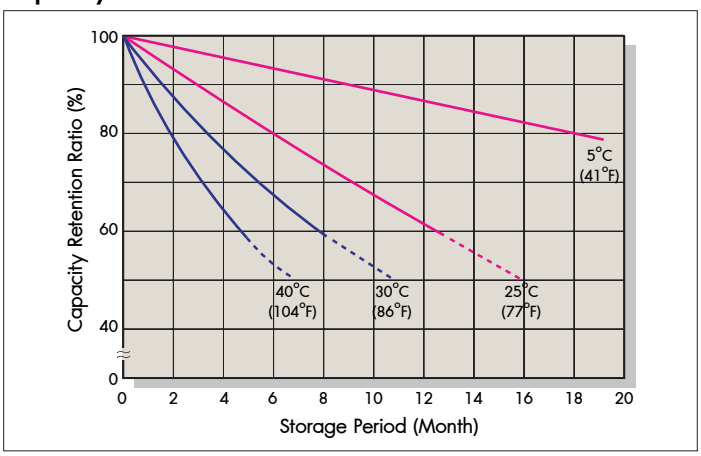
All mentioned values are average values

SL12-18 (12V 18Ah)

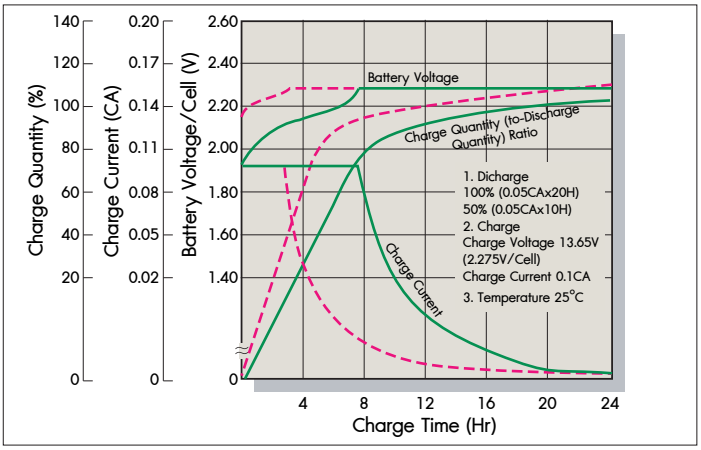
Trickle (or Float) Design Life



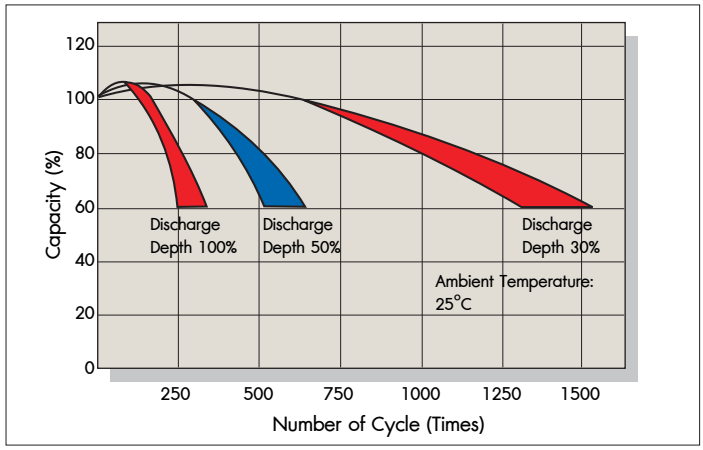
Capacity Retention Characteristic



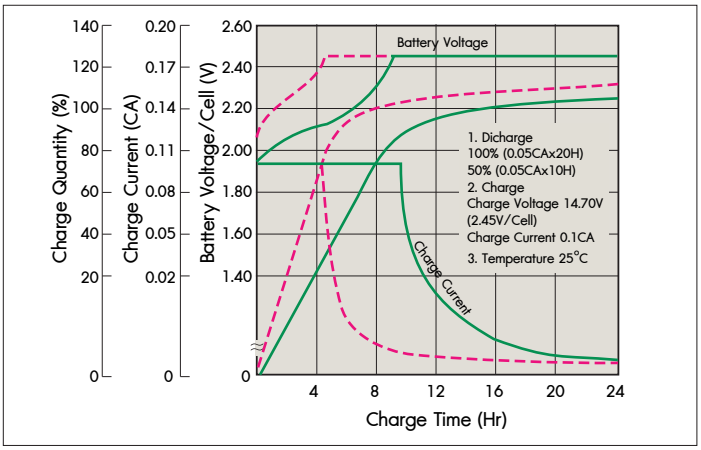
Battery Voltage and Charge Time for Standby Use



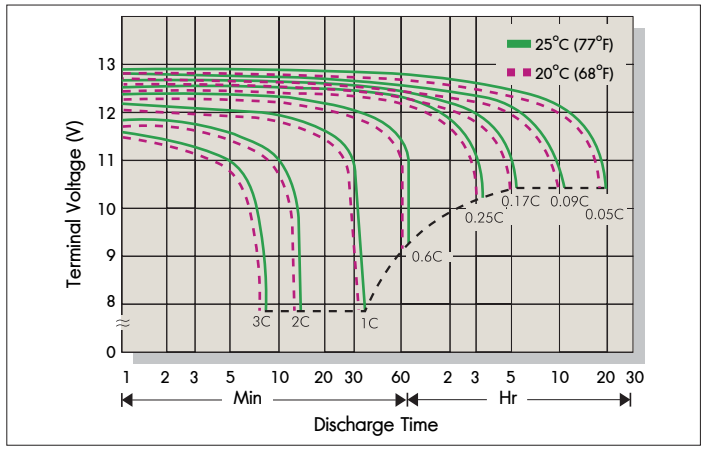
Cycle Service Life



Battery Voltage and Charge Time for Cycle Use



Terminal Voltage and Discharge Time



Charging Procedures

Application	Charge Voltage (V/cell)			Max. Charge Current
	Temperature	Set Point	Allowable Range	
Cycle Use	25°C	25°C	2.40~2.50	0.3C
Standby Use	25°C	25°C	2.25~2.30	

Discharge Current VS Discharge Voltage

Final Discharge Voltage (V/cell)	1.75	1.7	1.65	1.6
Discharge Current (A)	0.2C>(A)	0.2C<(A)<0.5C	0.2C<(A)<1C	(A)>1C

Effect of Temperature on Capacity (20HR)

Temperature	Dependency of Capacity (20HR)
40°C	102%
25°C	100%
0°C	85%
-15°C	65%

Self-discharge Characteristics

Charge Voltage (V/cell)	Preservation Rate
3 months	91%
6 months	82%
9 months	64%